CS5234 Mock Quiz 1

Solutions

Part A · Mock Quiz (20 pts)

Problem 1 (7 pts) · Estimating a sum

You have query access to an array $a_1, \ldots, a_n$ with every $a_i \in \{1, 2, \ldots, M\}$. Let $S = \sum_i a_i$.

(a) [2] Give an estimator X that uses one query and satisfies $\mathbb{E}[X] = S$.

Solution. Pick I uniformly from $[n]$, query a_I, output $X = n\,a_I$. Then $\mathbb{E}X = \sum_i \frac{1}{n} \cdot n a_i = S$.

(b) [3] Show $\mathrm{Var}(X) \le MS^2$. Conclude that averaging $k = O(M/\varepsilon^2)$ independent copies gives $\Pr[|\bar{X} - S| \le \varepsilon S] \ge 2/3$. *Hint: bound $\mathbb{E}[X^2]$ using $a_i \le M$, then use $S \ge n$.*

Solution. $\mathbb{E}X^2 = \sum_i \frac{1}{n} n^2 a_i^2 = n \sum_i a_i^2 \le n \sum_i M a_i = nMS$. Since every $a_i \ge 1$, $S \ge n$, so $\mathrm{Var}X \le \mathbb{E}X^2 \le nMS \le MS^2$. For the average $\bar{X}$ of k independent copies, $\mathbb{E}\bar{X} = S$ and $\mathrm{Var}\bar{X} \le MS^2/k$. Chebyshev: $\Pr[|\bar{X} - S| > \varepsilon S] \le \frac{MS^2/k}{\varepsilon^2 S^2} = \frac{M}{k\varepsilon^2}$. With $k = \lceil 3M/\varepsilon^2 \rceil$ this is $\le \frac{1}{3}$. *Marking: 1 pt for $\mathbb{E}X^2$, 1 pt for using $S \ge n$, 1 pt for Chebyshev + k.*

(c) [2] How do you boost the success probability to $1 - \delta$? What is the total number of queries?

Solution. Median trick: run the estimator of (b) independently t times and output the median of the t averages. The median is off by more than εS only if at least $t/2$ runs are off on the same side; each run fails w.p. $\le 1/3$, so by Chernoff this happens w.p. $\le 2e^{-t/100}$. Take $t = \lceil 100 \ln(2/\delta) \rceil = O(\log \frac{1}{\delta})$. Total queries $t \cdot k = O\big(\frac{M}{\varepsilon^2} \log \frac{1}{\delta}\big)$.

Problem 2 (7 pts) · Zero ones or k ones

The input $x \in \{0,1\}^n$ is promised to be either all zeros, or to contain exactly k ones ($1 \le k \le n/2$). Decide which, with success probability at least $2/3$.

(a) [2] Give a randomized algorithm with $O(n/k)$ queries and analyse it.

Solution. Query $t = \lceil 2n/k \rceil$ uniformly random positions (with replacement); answer "k ones" iff a 1 is seen. All-zero input: never sees a 1, always correct. k ones: each query hits a 1 w.p. k/n, so $\Pr[\text{miss all}] = (1 - \frac{k}{n})^t \le e^{-tk/n} \le e^{-2} < \frac{1}{3}$. One-sided error, $O(n/k)$ queries.

(b) [3] Let μ_0 be the all-zero string and μ_1 put ones on a uniformly random k-subset. For any deterministic algorithm making q queries, let ν_0, ν_1 be the distributions of its query–answer transcript under μ_0, μ_1. Prove $\Delta_{\mathrm{TV}}(\nu_0, \nu_1) \le qk/n$.

Solution. A deterministic algorithm that keeps receiving 0 follows one fixed path P that queries fixed positions $p_1, \ldots, p_q$; let τ_0 be that all-zero transcript. Under μ_0 the transcript is always τ_0, so ν_0 is a point mass. Under μ_1 the transcript differs from τ_0 only if some p_j lies in the random k-set; each fixed position is in it w.p. k/n, so by the union bound $\Pr_{\mu_1}[\text{transcript} \ne \tau_0] \le qk/n$. Hence $\Delta_{\mathrm{TV}}(\nu_0, \nu_1) = \frac{1}{2}\big[(1 - \nu_1(\tau_0)) + \sum_{\tau \ne \tau_0} \nu_1(\tau)\big] = 1 - \nu_1(\tau_0) \le \frac{qk}{n}$. *Key point: before the first 1 is seen, the queried positions do not depend on the input, which is why the union bound over fixed positions is valid.*

(c) [2] Use Yao's principle to conclude $R(f) = \Omega(n/k)$. What does this say about testing NotOR (take $k = \varepsilon n$)?

Solution. Take $\mu = \frac{1}{2}\mu_0 + \frac{1}{2}\mu_1$. Any deterministic algorithm with q queries distinguishes the two cases w.p. at most $\frac{1}{2} + \frac{1}{2}\Delta_{\mathrm{TV}}(\nu_0, \nu_1) \le \frac{1}{2} + \frac{qk}{2n}$. Being correct w.p. $\ge \frac{2}{3}$ needs $q \ge \frac{n}{3k}$, so $D_\mu(f) \ge \frac{n}{3k}$ and by Yao $R(f) \ge D_\mu(f) = \Omega(n/k)$; with (a), $R(f) = \Theta(n/k)$. For NotOR with $k = \varepsilon n$: strings with εn ones are ε-far from all-zero, so every tester must solve this problem, giving $\Omega(1/\varepsilon)$ queries, which matches the $O(1/\varepsilon)$ tester.

Problem 3 (6 pts) · Streaming sample of size k

A stream $x_1, \dots, x_n \in [m]$ arrives one element at a time; n is unknown in advance. Algorithm: store the first k elements; when the i-th element arrives ($i > k$), with probability k/i it replaces a *uniformly random* one of the k stored elements.

(a) [3] Prove by induction that after processing element i ($i \geq k$), every x_j with $j \leq i$ is in the sample with probability exactly k/i.

Solution. *Base* $i = k$: all k elements are stored, probability $1 = k/k$. *Step*: assume the claim after step $i-1$. The new element x_i enters w.p. k/i. For $j < i$, x_j is in the sample after step i iff it was there after step $i-1$ (prob. $\frac{k}{i-1}$) and was not evicted; eviction needs x_i accepted ($\frac{k}{i}$) and x_j's slot chosen ($\frac{1}{k}$), independent of the past. So $\Pr = \frac{k}{i-1}\left(1 - \frac{k}{i} \cdot \frac{1}{k}\right) = \frac{k}{i-1} \cdot \frac{i-1}{i} = \frac{k}{i}$.

(b) [1] How much space does the algorithm use?

Solution. k stored values of $\log m$ bits plus a counter i of $\log n$ bits: $O(k \log m + \log n)$ bits.

(c) [2] Let p be the fraction of even numbers in the stream, estimated by the fraction $\hat{p}$ of even numbers in the final sample. You may use that the indicators "x_j is in the sample" are pairwise non-positively correlated. Show that $k = O(1/\varepsilon^2)$ gives $\Pr[|\hat{p} - p| \leq \varepsilon] \geq 2/3$.

Solution. Let E be the positions holding even numbers, $|E| = pn$, and $Y_j = \mathbf{1}[x_j \in \text{sample}]$. Then $\hat{p} = \frac{1}{k}\sum_{j \in E} Y_j$ and by (a) $\mathbb{E}\hat{p} = \frac{1}{k} \cdot pn \cdot \frac{k}{n} = p$. Non-positive covariances give $\mathrm{Var}(\sum_{j \in E} Y_j) \leq \sum_{j \in E} \mathrm{Var} Y_j \leq \sum_{j \in E} \mathbb{E} Y_j = pk$, so $\mathrm{Var}\hat{p} \leq \frac{p}{k} \leq \frac{1}{k}$. Chebyshev: $\Pr[|\hat{p} - p| > \varepsilon] \leq \frac{1}{k\varepsilon^2} \leq \frac{1}{3}$ for $k = \lceil 3/\varepsilon^2 \rceil$. *(Chernoff is not directly allowed: the Y_j are not independent.)*

Part B · Extra practice

E1 · Maximum load (L1)

Throw n balls independently and uniformly into n bins. (a) Show that with probability $1 - o(1)$ no bin receives more than $10 \ln n$ balls. (b) Why does Chebyshev's inequality not suffice here?

Solution. (a) Fix bin j; its load $X = \sum_i X_i$ with independent $X_i \in \{0, 1\}$ and $\mu = 1$. Chernoff ($\delta > 1$) with $1 + \delta = 10 \ln n$: $\Pr[X \geq 10 \ln n] \leq e^{-\delta\mu/3} = e^{-(10 \ln n - 1)/3} = e^{1/3} n^{-10/3}$. Union bound over n bins: $\leq e^{1/3} n^{-7/3} = o(1)$. (b) $\mathrm{Var} X \leq 1$, so Chebyshev only gives $\Pr[X \geq 10 \ln n] \leq \frac{1}{(10 \ln n - 1)^2}$ per bin; multiplied by n bins the union bound exceeds 1. The union bound needs a per-bin tail smaller than $1/n$, which only an exponential (Chernoff) tail provides.

E2 · Counting with relative error (L2)

An n-bit string has A ones, and you are promised $A \geq t$ for a known t. (a) Give a $(1 \pm \varepsilon)$-approximation of A with probability $2/3$ using $O\left(\frac{n}{t\varepsilon^2}\right)$ queries. (b) Sketch how to proceed if t is unknown (but $A \geq 1$).

Solution. (a) Sample k positions with replacement, output $\hat{A} = X\frac{n}{k}$. $\mu = \mathbb{E}X = \frac{kA}{n} \geq \frac{kt}{n}$. Chernoff with $\delta = \varepsilon$: $\Pr[|X - \mu| \geq \varepsilon\mu] \leq 2e^{-\varepsilon^2\mu/3} \leq 2e^{-\varepsilon^2 kt/(3n)}$. With $k = \frac{6n}{t\varepsilon^2}$ this is $2e^{-2} < \frac{1}{3}$. (b) Guess $t' = n/2, n/4, \dots$ as in the edge-counting algorithm: run (a) with t' and stop the first time the estimate is at least $2t'$. While t' is much larger than A, the estimate stays small (upper tail of Chernoff with $\mu_H = kt'/n$); once $t' \leq A$ the estimate is accurate. The costs form a geometric series dominated by the last round, $O(\frac{n}{A\varepsilon^2})$; boost each round with the median trick and apply a union bound over the $O(\log n)$ rounds.

E3 · Testing whether a string is constant (L4)

Property: all bits of $x \in \{0, 1\}^n$ are equal. (a) What is the distance of x from the property? (b) Give a one-sided tester with $O(1/\varepsilon)$ queries and prove it works.

Solution. (a) $\min(\#0, \#1)$: change the minority bits. So ε-far ⇔ the minority fraction $m \geq \varepsilon$ (with $m \leq \frac{1}{2}$). (b) Sample $k = \lceil 2/\varepsilon \rceil$ positions; accept iff all sampled bits are equal. A constant string is always accepted. If x is ε-far, all samples equal requires all-majority or all-minority: Pr $= (1-m)^k + m^k \leq 2(1-m)^k \leq 2(1-\varepsilon)^k \leq 2e^{-2} < \frac{1}{3}$ (using $m \leq 1-m$).

E4 · Triangle detection needs $\Omega(n^2)$ (L3)

With pair queries on an n-vertex graph, prove that deciding whether the graph contains a triangle has randomized query complexity $\Omega(n^2)$. *Hint: reduce from OR; add one special vertex.*

Solution. Given an OR instance with $N = \binom{n-1}{2}$ bits indexed by pairs $\{i,j\} \subseteq [n-1]$, build a graph on vertices $w, a_1, \ldots, a_{n-1}$: w is adjacent to every a_i, and $a_i a_j$ is an edge iff bit $\{i,j\}$ is 1. *Correctness*: if bit $\{i,j\} = 1$ then w, a_i, a_j is a triangle; if all bits are 0 the graph is a star, which has no triangle. *Simulation*: a query (w, a_i) is answered 1 for free; a query (a_i, a_j) costs one OR query ($C = 1$). Hence $R(\text{Triangle}) \geq R(\text{OR}_N) \geq N/3 = \Omega(n^2)$.